WiFi Controlled 4WD RC Car with Live Camera and Automatic Lights

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• Abstract

This project presents the design and development of a WiFi controlled RC car using ESP8266 and ESP32-CAM modules. The system allows wireless control of the vehicle using a web interface connected through an IP address. The vehicle is designed with a 4-wheel drive mechanism powered by DC motors and controlled through an L298N motor driver. A steering motor is used to control turning of the vehicle.

In addition to motion control, the system includes an ESP32-CAM module that provides live video streaming and smooth controlling up to approximately 10–15 meters. The project also integrates photo sensors (LDR) that automatically switch on LED headlights in low-light conditions. The project demonstrates concepts of robotics, wireless communication, embedded systems and Internet of Things (IoT).

• Introduction

Remote controlled robotic vehicles are widely used in modern technology applications such as surveillance, exploration, rescue operations and military systems. With the development of microcontrollers and wireless communication modules, it is possible to build intelligent robotic systems at low cost. This project focuses on developing a WiFi controlled RC car that can be controlled remotely through a smartphone or computer using an IP address. The system integrates mechanical design, electronics and embedded programming. The ESP8266 microcontroller is used for motor control and communication while an ESP32-CAM module is used to provide real-time video streaming.

The integration of automatic lighting using photo sensors enhances the system by enabling the vehicle to operate effectively even in low light environments.

· Objectives

- Design and build a WiFi controlled robotic vehicle.
- Implement a 4-wheel drive power transmission mechanism.
- Integrate an ESP32-CAM module for live video monitoring.
- Implement automatic headlight control using photo sensors.
- Understand the working of motor drivers and embedded control systems.

· Components Used

1 .ESP8266 (ESP-12E) – WiFi microcontroller used for controlling the motors
2. ESP32-CAM – Camera module for video streaming
3. L298N Motor Driver – Used to drive the DC motors
4. DC Motors – Used for movement and steering
5. Wheels – Four wheels for vehicle movement
6.Gears – Used to distribute power for 4-wheel drive
7. LDR Photo Sensors – Detect light intensity
8. LED Lights – Used as headlights
9. Battery – Power supply
10. Chassis – Mechanical body of the car
11. buck converter - to step down the voltage

· Mechanical Design

The mechanical structure of the RC car is designed using a chassis with four wheels. Two DC motors are used for driving the vehicle. Using a gear mechanism, the power from these motors is transferred to all four wheels, enabling a four-wheel drive system. This increases traction and stability.

A separate motor is used for the steering mechanism which allows the vehicle to turn left or right. The use of gears ensures efficient power transmission while maintaining mechanical simplicity.

• System Architecture

The system consists of two main subsystems: the control system and the camera system.

The control system is based on the ESP8266 microcontroller which receives commands from the user through WiFi. These commands are processed and sent to the L298N motor driver which controls the motors.

The camera system uses an ESP32-CAM module which creates a separate WiFi network and provides live video streaming through a unique IP address.

The lighting system uses photo sensors to detect ambient light levels and automatically switch on the LED headlights when the environment becomes dark.

• Working Principle

1. The user connects to the WiFi network created by the ESP8266 module.
2. A control webpage is opened using the IP address of the module.
3. Commands such as forward, reverse, left and right are sent from the user device.
4. The ESP8266 processes the commands and activates the L298N motor driver.
5. The motor driver powers the motors accordingly to move the vehicle.
6. The ESP32-CAM module streams live video to the user.
7. The LDR sensor monitors ambient light and automatically turns on LED headlights when necessary.

• Applications

- Surveillance robots

- Military monitoring vehicles
- Disaster rescue robots
- Remote inspection vehicles
- Educational robotics projects

• Advantages

- Wireless control using WiFi
- Real-time video monitoring
- Automatic headlight system
- Low cost and easy to build
- Expandable with additional sensors

• Limitations

- Limited WiFi range
- Battery life limitations
- Camera streaming delay at long distance

• Future Improvements

- Add obstacle detection sensors
- Implement autonomous navigation
- Increase camera range and quality
- Add GPS tracking
- Integrate AI-based object detection

• Suspension System Characteristics and Off-Road Performance

The RC car is equipped with a spring–damper suspension system designed to absorb shocks and maintain wheel contact with the ground while the vehicle moves across uneven terrain. The suspension mainly consists of coil springs and shock absorbers connected between the chassis and the wheel assemblies. This system plays a critical role in maintaining stability and controlling vibrations generated during movement.

During testing, it was observed that the suspension has relatively high stiffness compared to the total mass of the vehicle. Suspension stiffness refers to the resistance of the spring to compression under an applied load. Due to the higher stiffness, the suspension compresses very little during normal operation, resulting in limited suspension travel.

In the context of off-road driving, an effective suspension system is very important because the vehicle frequently encounters bumps, obstacles, and irregular surfaces such as gravel, small rocks, or uneven ground. A properly tuned suspension allows the wheels to move vertically and adapt to the terrain, helping the tires maintain consistent contact with the surface. This improves traction, stability, and control of the RC car during off-road operation.

However, when the suspension is too stiff, the wheels may not adapt efficiently to rough terrain, which can reduce shock absorption and affect the overall off-road capability of the vehicle. For better off-road performance, the suspension system should allow adequate compression and rebound, enabling the vehicle to absorb impacts while maintaining chassis stability.

To improve off-road performance, adjustments such as using springs with a lower spring constant, optimizing shock absorber positioning, or modifying the vehicle’s weight distribution can be considered. Proper suspension tuning ensures better terrain adaptability, smoother movement over obstacles, and improved overall performance of the RC car in off-road conditions.

Conclusion

The WiFi controlled RC car with live camera and automatic lighting demonstrates the integration of robotics, wireless communication and embedded systems. The project successfully shows how a robotic vehicle can be controlled remotely while providing real-time monitoring. The addition of automatic lighting and a camera module enhances the usability of the vehicle in real-world applications. The project also serves as a learning platform for robotics, IoT and electronics engineering.